A Periodic Damping Family with Identical Fixed Points and Different Convergent Message Trajectories

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Abstract

Conjecture 00000004530 is true. We give an explicit family of synchronous message-passing algorithms on a graph consisting of one edge. The damping schedule has one averaging step every p updates, so its averaging frequency is $1/p$. Every member has the same fixed-point set and reaches consensus in at most p updates. The cases $p = 1$ and $p = 2$ have different trajectories from the same initial messages. All claims needed for the construction are verified in a standalone Lean 4 project.

1 Statement and interpretation

The English statement of Conjecture 00000004530 is:

> There exist two algorithms with the same fixed-point set but different convergence trajectories, and the separation is constructed by an explicit frequency-sweep family of damping parameters.

The Chinese statement has the same existential content. Neither version specifies a particular message alphabet, graph, or continuous frequency range. We supply a discrete frequency sweep: the period p runs over all positive integers, and the averaging frequency runs over $1, 1/2, 1/3, \ldots$. For a time-dependent algorithm, a fixed state means a state left unchanged by every scheduled update. In the construction below, this also equals the fixed-point set of each individual update and of the one-period update. No uniqueness of the limit, or different limiting values, is asserted by the conjecture: it asks for different convergence trajectories.

2 The message-passing family

Take two vertices joined by an edge, and let a, b be their two directed messages. The undamped synchronous update copies the incoming message:

$$F(a, b) = (b, a).$$

Use the invariant message space

$$\mathcal{M} = \{(c + d, c - d) : c, d \in \mathbb{Z}\}.$$

Equivalently, a, b are integers of equal parity. The coordinates (c, d) are unique: $c = (a + b)/2$ and $d = (a - b)/2$. Thus the two-node update is an ordinary local message-passing operation, and the coordinate representation is injective.

For each integer $p \geq 1$, define, for update indices $n \geq 0$,

$$\alpha_n^{(p)} = \begin{cases} 1/2, & p \mid n + 1, \\ 1, & p \nmid n + 1. \end{cases}$$

The algorithm A_p performs the usual damped update

$$(a_{n+1}, b_{n+1}) = (1 - \alpha_n^{(p)})(a_n, b_n) + \alpha_n^{(p)}(b_n, a_n). \tag{1}$$

All coefficients are valid damping parameters in $(0, 1]$. At a full update, (a, b) is swapped. At a half update, both messages become $(a + b)/2$, which is an integer because the input lies in $\mathcal{M}$; the output again lies in $\mathcal{M}$. There is no integer rounding.

The schedule is periodic with period p. Its half-damped updates occur exactly at $n = p - 1, 2p - 1, 3p - 1, \ldots$. For $p \geq 2$ the period is fundamental: a smaller positive period would have to send the half-damped index $p - 1$ to a non-half-damped index. Consequently there is exactly one half update per p updates. Varying p gives the promised explicit frequency sweep.

Lemma 1. *For every $p \geq 1$ the fixed-point set of A_p is*

$$\mathrm{Fix}(A_p) = \{(c, c) : c \in \mathbb{Z}\}.$$

Every individual scheduled update has this same fixed-point set.

Proof. For a swap, $(b, a) = (a, b)$ if and only if $a = b$. For an averaging step, $((a+b)/2, (a+b)/2) = (a, b)$ if and only if $a = b$. Both directions hold on $\mathcal{M}$. Therefore a message pair is unchanged by every scheduled update if and only if it is a consensus pair (c, c). □

Lemma 2. *Starting from $(a_0, b_0) = (c + d, c - d)$, algorithm A_p satisfies*

$$(a_n, b_n) = \begin{cases} (c + (-1)^n d, c - (-1)^n d), & 0 \leq n < p, \\ (c, c), & n \geq p. \end{cases}$$

In particular every trajectory converges, and it reaches its limit after at most p updates.

Proof. In coordinates, the average c is preserved. Each full update sends d to $-d$, and each half update sends d to 0. The first $p - 1$ updates are full; update p is half. Subsequent updates preserve $d = 0$. These statements prove the displayed formula by induction. Eventual equality to (c, c) implies convergence in the usual Euclidean metric, as well as in the discrete topology on $\mathcal{M}$. □

The one-period update therefore maps every $(c + d, c - d)$ to (c, c), so its fixed points are also exactly those of Lemma 1.

Theorem 1. *The algorithms A_1 and A_2 prove Conjecture 00000004530.*

Proof. They are two members of the explicit damping-frequency family above. Their fixed-point sets are identical by Lemma 1, and all of their trajectories converge by Lemma 2. From the same initial messages $(1, -1)$ their trajectories begin

	$n = 0$	$n = 1$	$n = 2$	$n \geq 3$
A_1	$(1, -1)$	$(0, 0)$	$(0, 0)$	$(0, 0)$
A_2	$(1, -1)$	$(-1, 1)$	$(0, 0)$	$(0, 0)$.

The states at $n = 1$ are different, which proves the required separation of convergence trajectories. □

3 Lean verification and correspondence

The project uses Lean 4.24.0 and imports only `Std`. The entry point is `lean/Main.lean`; from the `lean` directory run `lake build`. The proof contains no proof placeholders, externally trusted computation, or additional axioms.

Mathematical object or assertion	Lean declaration
$2\alpha_n^{(p)}$	`twiceDamping`
Coordinate change $(c, d) \mapsto (c + d, c - d)$	`messages`
Injectivity of coordinates	`messages_injective`
Exact damped local update (1)	`message_passing_identity`
Periodicity and reset locations	`damping_periodic`, `first_reset`, `before_reset`
Fixed-point set	`state_fixed_iff`
Actual iteration recurrence	`trajectory_recurrence`
Finite-time consensus for every initial state	`consensus`
Existential separation	`conjecture4530`

The damping identity is denominator-cleared by multiplication by 2; its right-hand side uses the physical messages themselves. It therefore verifies genuine convex damping, not merely an unrelated scalar recurrence. The main theorem quantifies over every integer coordinate state, proves agreement of the fixed sets, proves eventual equality to consensus for both algorithms, and proves inequality of their physical messages after one update from the common initial state. The printed axiom audit lists only Lean's standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound`.